

Bo Li

CONTACT INFORMATION	Department of Computer Science University of Illinois at UrbanaChampaign Address: 801 South Wright Street Champaign, IL USA 61820 Homepage: http://boli.cs.illinois.edu/	<i>Mobile:</i> 615-315-1158 <i>e-mail:</i> lbo@illinois.edu
RESEARCH INTERESTS	Machine learning, security, privacy, trustworthy machine learning, algorithmic game theory, data mining, social networks	
EDUCATION	Vanderbilt University, Nashville, TN, USA PhD, Computer Science Sept. 2011 to Aug. 2016 Tongji University, Shanghai, China Bachelor, Computer Science, Sept. 2007 to July 2011	
PROFESSIONAL EXPERIENCE	University of Illinois at UrbanaChampaign, <i>Assistant Professor</i> Aug. 2018 - present University of California, Berkeley, <i>Postdoctoral Researcher</i> Jun. 2016 - Aug. 2018 Advisors: Dawn Song	
HONORS AND AWARDS	• Alfred P. Sloan Research Fellowship 2022 • Facebook Research Award 2021 • NSF CAREER Award 2020 • Amazon Research Award 2020 • MIT Technology Review list of 35 Innovators Under 35, 2020 • Intel Rising Star Award 2020 • Amazon Research Award 2019 • Facebook Research Award 2019 • IBM Research Award 2019 • Best paper at International Conference on Embedded Wireless Systems and Networks (EWSN), 2021 • Best paper at IEEE/ACM International Conference on Automated Software Engineering (ASE), 2018 • Best paper at ACM Workshop on Artificial Intelligence and Security (AISec), 2017 • Best paper at Deep Learning and Security Workshop (DLSRF), 2017 • Symantec Research Labs Graduate Fellowship 2015 (one of the three recipients Internationally) • IEEE Symposium on Security and Privacy 2015 Scholarship • National Scholarship (1%) (2010) • First Place Prize Scholarship (2008-2010) • Outstanding Student Prize (2008-2010) • ACM/ICPC Asia Regional Contest, Second Place, March 2009 • One Hundred Distinguish Students in Shanghai (2008-2009) • Outstanding students leader prize (2008-2010) • Best volunteer prize for EXPO 2010 (2010) • Travel awards for NIPS, ICDM, ICML, AISTATS, GREPSEC	
PUBLICATIONS	[1] Xinlei Pan, Chaowei Xiao, Warren He, Shuang Yang, Jian Peng, Mingjie Sun, Mingyan Liu, <u>Bo Li</u> and Dawn Song. <i>Characterizing Attacks on Deep Reinforcement Learning</i>. AAMAS 2022 (Oral presentation).	

- [2] Boxin Wang*, Chejian Xu*, Shuohang Wang, Zhe Gan, Yu Cheng, Jianfeng Gao, Ahmed Hassan Awadallah, Bo Li. *Adversarial GLUE: A Multi-Task Benchmark for Robustness Evaluation of Language Models*. **NeurIPS** 2021 (Oral presentation).
- [3] Dan Hendrycks, Mantas Mazeika, Andy Zou, Sahil Patel, Christine Zhu, Jesus Navarro, Dawn Song, Bo Li, Jacob Steinhardt. *What Would Jiminy Cricket Do? Towards Agents That Behave Morally*. **NeurIPS** 2021.
- [4] Yunhui Long*, Boxin Wang*, Zhuolin Yang, Bhavya Kailkhura, Aston Zhang, Carl A. Gunter, Bo Li. *G-PATE: Scalable Differentially Private Data Generator via Private Aggregation of Teacher Discriminators*. **NeurIPS** 2021.
- [5] Zhuolin Yang, Linyi Li, Xiaojun Xu, Shiliang Zuo, Qian Chen, Pan Zhou, Benjamin I. P. Rubinstein, Ce Zhang, Bo Li. *TRS: Transferability Reduced Ensemble via Promoting Gradient Diversity and Model Smoothness*. **NeurIPS** 2021.
- [6] Jingkang Wang, Tianyun Zhang, Sijia Liu, Pin-Yu Chen, Jiachen Xu, Makan Fardad, Bo Li. *Adversarial Attack Generation Empowered by Min-Max Optimization*. **NeurIPS** 2021.
- [7] Yige Li, Xixiang Lyu, Xingjun Ma, Nodens Koren, Lingjuan Lyu, Bo Li. *Anti-Backdoor Learning: Training Clean Models on Poisoned Data*. **NeurIPS** 2021.
- [8] Fan Wu, Yunhui Long, Ce Zhang, Bo Li. *LinkTeller: Recovering Private Edges from Graph Neural Networks via Influence Analysis*. IEEE Symposium on Security and Privacy (**IEEE S&P**), 2021.
- [9] Hengtong Zhang, Tianhang Zheng, Yaliang Li, Jing Gao, Lu Su and Bo Li. *Profanity-Avoiding Training Framework for Seq2seq Models with Certified Robustness*. **EMNLP** 2021.
- [10] Mingjie Sun, Zichao Li, Chaowei Xiao, Haonan Qiu, Bhavya Kailkhura, Mingyan Liu, Bo Li. *Can Shape Structure Features Improve Model Robustness?*. **ICCV** 2021.
- [11] Chulin Xie, Pin-Yu Chen, Minghao Chen, Bo Li. *CRFL: Certifiably Robust Federated Learning against Backdoor Attacks*. **ICML** 2021.
- [12] Jiawei Zhang, Linyi Li, Huichen Li, Xiaolu Zhang, Shuang Yang, Bo Li. *Progressive-Scale Boundary Blackbox Attack via Projective Gradient Estimation*. **ICML** 2021.
- [13] Kaizhao Liang, Jacky Zhang, Boxin Wang, Zhuolin Yang, Sanmi Koyejo, Bo Li. *Uncovering the Connections Between Adversarial Transferability and Knowledge Transferability*. **ICML** 2021.
- [14] Haoxiang Wang, Han Zhao, Bo Li. *Bridging Multi-Task Learning and Meta-Learning: Towards Efficient Training and Effective Adaptation*. **ICML** 2021.
- [15] Nezihe Merve Grel, Xiangyu Qi, Luka Rimanic, Ce Zhang, Bo Li. *Knowledge Enhanced Machine Learning Pipeline against Diverse Adversarial Attacks*. **ICML** 2021.
- [16] Boxin Wang, Fan Wu, Yunhui Long, Luka Rimanic, Ce Zhang, Bo Li. *DataLens: Scalable Privacy Preserving Training via Gradient Compression and Aggregation*. **CCS** 2021.
- [17] Linyi Li*, Maurice Weber*, Xiaojun Xu, Luka Rimanic, Bhavya Kailkhura, Tao Xie, Ce Zhang, Bo Li. *TSS: Transformation-Specific Smoothing for Robustness Certification*. **CCS** 2021.

- [18] Yulong Cao, Ningfei Wang, Chaowei Xiao, Dawei Yang, Jin Fang, Ruigang Yang, Qi Alfred Chen, Mingyan Liu, Bo Li. *Security of Multi-Sensor Fusion based Perception in Autonomous Driving Under Physical-World Attacks*. IEEE Symposium on Security and Privacy (**IEEE S&P**), 2021.
- [19] Aria Rezaei, Chaowei Xiao, Jie Gao, Bo Li, Sirajum Munir. *Application-Driven Privacy-Preserving Data Publishing with Correlated Attributes*. Embedded Wireless Systems and Networks (**EWSN 2021**) (**Best Paper Award**)
- [20] Ruoxi Jia, Fan Wu, Xuehui Sun, Jiachen Xu, David Dao, Bhavya Kailkhura, Ce Zhang, Bo Li, Dawn Song. *Scalability vs. Utility: Do We Have to Sacrifice One for the Other in Data Importance Quantification?*. **CVPR** 2021.
- [21] Maurice Weber, Nana Liu, Bo Li, Ce Zhang, Zhikuan Zhao. *Optimal Provable Robustness of Quantum Classification via Quantum Hypothesis Testing*. **npj Quantum Information** 2021.
- [22] Xinyun Chen, Wenxiao Wang, Chris Bender, Yiming Ding, Ruoxi, Jia, Bo Li, Dawn Song. *REFIT: a Unified Watermark Removal Framework for Deep Learning Systems with Limited Data*. **CCS** 2021.
- [23] Huichen Li*, Linyi Li*, Xiaojun Xu, Xiaolu Zhang, Shuang Yang, Bo Li. *Nonlinear Projection Based Gradient Estimation for Query Efficient Blackbox Attacks*. **AISTATS** 2021.
- [24] Zhuolin Yang, Zhaoxi Chen, Tiffany Cai, Xinyun Chen, Bo Li, Yuandong Tian. *Understanding Robustness in Teacher-Student Setting: A New Perspective*. **AISTATS** 2021.
- [25] Wenhao Ding, Baiming Chen, Bo Li, Ji Eun Kim, Ding Zhao. *Multimodal Safety-Critical Scenarios Generation for Decision-Making Algorithms Evaluation*. **ICRA** 2021.
- [26] Boxin Wang, Shuohang Wang, Yu Cheng, Zhe Gan, Ruoxi Jia, Bo Li, Jingjing Liu. *InfoBERT: Improving Robustness of Language Models from An Information Theoretic Perspective*. **ICLR** 2021.
- [27] Yige Li, Xingjun Ma, Nodens Koren, Lingjuan Lyu, Xixiang Lyu, Bo Li. *Neural Attention Distillation: Erasing Backdoor Triggers from Deep Neural Networks*. **ICLR** 2021.
- [28] Wenhao Ding, Baiming Chen, Bo Li, Ji Eun Kim, Ding Zhao. *Multimodal Safety-Critical Scenarios Generation for Decision-Making Algorithms Evaluation*. IEEE Robotics and Automation Letters (**RA-L**) 2021.
- [29] Luka Rimanic, Cedric Renggli, Bo Li, Ce Zhang. *On Convergence of Nearest Neighbor Classifiers over Feature Transformations*. **NeurIPS** 2020.
- [30] Huan Zhang, Hongge Chen, Chaowei Xiao, Bo Li, Mingyan Liu, Duane Boning, Cho-Jui Hsieh. *Robust Deep Reinforcement Learning against Adversarial Perturbations on State Observations*. **NeurIPS** 2020.
- [31] Boxin Wang, Hengzhi Pei, Boyuan Pan, Qian Chen, Shuohang Wang, Bo Li. *T3: Tree-Autoencoder Regularized Adversarial Text Generation for Targeted Attack*. **EMNLP** 2020.
- [32] Xiaojun Xu*, Qi Wang, Huichen Li, Nikita Borisov, Carl A. Gunter, Bo Li. *Detecting AI Trojans Using Meta Neural Analysis*. IEEE Symposium on Security and Privacy (**IEEE S&P**), 2020.

- [33] Shawn Shan*, Emily Wenger, Bolun Wang, Bo Li, Haitao Zheng, Ben Y. Zhao. *Gotta Catch ‘Em All: Using Honeypots to Catch Adversarial Attacks on Neural Networks*. **CCS** 2020.
- [34] Haonan Qiu*, Chaowei Xiao*, Lei Yang, Xinchun Yan, Honglak Lee, Bo Li. *SemanticAdv: Generating Adversarial Examples via Attribute-conditioned Image Editing*. **ECCV** 2020.
- [35] Boyuan Pan*, Yazheng Yang, Kaizhao Liang*, Bhavya Kailkhura, Zhongming Jin, Xian-Sheng Hua, Deng Cai, Bo Li, *Adversarial Mutual Information for Text Generation*. **ICML** 2020.
- [36] Ankit Raj*, Yoram Bresler, Bo Li, *Improving Robustness of Deep-Learning-Based Image Reconstruction*. **ICML** 2020.
- [37] Gerald Friedland, Jingkang Wang*, Ruoxi Jia, Bo Li, *The Helmholtz Method: Using Perceptual Compression to Reduce Machine Learning Complexity*. (**IEEE on Multimedia Information Processing and Retrieval** 2020).
- [38] Zhenkai Zhang, Zihao Zhan*, Daniel Balasubramanian, Bo Li, Peter Volgyesi, Xenofon Koutsoukos, *Leveraging EM Side-Channel Information to Detect Rowhammer Attacks*. IEEE Symposium on Security and Privacy (**IEEE S&P**), 2020.
- [39] Huichen Li*, Xiaojun Xu*, Xiaolu Zhang, Shuang Yang, Le Song, Yuan Qi, Bo Li, *QEBA: Query-Efficient Boundary-Based Blackbox Attack*. **CVPR** 2020.
- [40] Yuheng Zhang*, Ruoxi Jia*, Hengzhi Pei*, Wenxiao Wang*, Bo Li, Dawn Song, *Secret Revealer: Generative Model-Inversion Attacks Against Deep Neural Networks*. **CVPR** 2020.
- [41] Lei Huang*, Li Liu, Fan Zhu, Diwen Wan, Zehuan Yuan, Bo Li, Ling Shao, *Controllable Orthogonalization in Training DNNs*. **CVPR** 2020. (Finalist of Best Paper)
- [42] Chao Yan*, Haifeng Xu, Yevgeniy Vorobeychik, Bo Li, Daniel Fabbri, Bradley Malin, *To Warn or Not to Warn: Online Signaling in Audit Games*. **ICDE** 2020.
- [43] Huan Zhang, Hongge Chen, Chaowei Xiao*, Sven Gowal, Robert Stanforth, Bo Li, Duane Boning, Cho-Jui Hsieh, *Towards Stable and Efficient Training of Verifiably Robust Neural Networks*. **ICLR** 2020.
- [44] Anand Bhattad*, Min Jin Chong*, Kaizhao Liang*, Bo Li, David Forsyth, *Unrestricted Adversarial Examples via Semantic Manipulation*. **ICLR** 2020.
- [45] Chulin Xie*, Keli Huang*, Pin-Yu Chen, Bo Li, *DBA: Distributed Backdoor Attacks against Federated Learning*. **ICLR** 2020.
- [46] Yuyu Zhang, Xinshi Chen, Yuan Yang, Arun Ramamurthy, Bo Li, Yuan Qi, Le Song, *Efficient Probabilistic Logic Reasoning with Graph Neural Networks*. **ICLR** 2020.
- [47] Yunan Ye*, Hengzhi Pei*, Boxin Wang*, Pin-Yu Chen, Yada Zhu, Bo Li, *Reinforcement-Learning based Portfolio Management with Augmented Asset Movement Prediction States*. **AAAI** 2020.
- [48] Jingkang Wang*, Yang Liu, Bo Li, *Reinforcement Learning with Perturbed Rewards*. **AAAI** 2020.
- [49] Saikiran Bulusu, Bhavya Kailkhura, Bo Li, Pramod K. Varshney, Dawn Song. *Anomalous Example Detection in Deep Learning: A Survey*. **IEEE Access** 2020.

- [50] Ruoxi Jia*, David Dao, Boxin Wang*, Frances Ann Hubis, Nezihe Merve Gurel, Bo Li, Ce Zhang, Costas Spanos, Dawn Song, *Efficient Task-Specific Data Valuation for Nearest Neighbor Algorithms*. **PVLDB** 2019.
- [51] Kin Sum Liu, Bo Li, Chaowei Xiao*, Jie, Gao, *Performing Co-Membership Attacks Against Deep Generative Models*. **ICDM** 2019.
- [52] Ruizhi Deng*, Chaowei Xiao*, Bo Li, *Identifying adversarial videos*. **ICCV** 2019.
- [53] Liang Tong, Bo Li, Chen Hajaj, Chaowei Xiao*, Ning Zhang, Yevgeniy Vorobeychik, *Improving Robustness of ML Classifiers against Realizable Evasion Attacks Using Conserved Features*. **USENIX Security**, 2019.
- [54] Linyi Li*, Zexuan Zhong*, Tao Xie, Bo Li. *Robustra: Training Provable Robust Neural Networks over Reference Adversarial Space*. **IJCAI** 2019. (Oral Presentation)
- [55] Kimin Lee*, Sukmin Yun, Kibok Lee, Honglak Lee, Bo Li, Jinwoo Shin. *Robust Inference via Generative Classifiers for Handling Noisy Labels*. **ICML** 2019.
- [56] Chong Xiang*, Charles R. Qi, Bo Li. *Generating 3D Adversarial Point Clouds*. **CVPR** 2019.
- [57] Chaowei Xiao*, Dawei Yang, Bo Li, Jia Deng, and Mingyan Liu. *Realistic adversarial examples in 3d meshes*. **CVPR** 2019. (Oral Presentation)
- [58] Ruoxi Jia*, David Dao, Boxin Wang*, Frances Ann Hubis, Nick Hynes, Bo Li, Ce Zhang, Dawn Song, Costas Spanos. *Towards Efficient Data Valuation Based on the Shapley Value*. **AISTAS** 2019
- [59] Zhuolin Yang*, Bo Li, Pin-Yu Chen, Dawn Song. *Characterizing Audio Adversarial Examples Using Temporal Dependency*. **ICLR** 2019
- [60] Jinfeng Li*, Shouling Ji, Tianyu Du, Bo Li, Ting Wang. *TextBugger: Generating Adversarial Text Against Real-world Applications*. **NDSS** 2019.
- [61] Xiang Ling*, Shouling Ji, Jiaxu Zou, Jiannan Wang, Chunming Wu, Bo Li, Ting Wang. *DeepSec: A Uniform Platform for Security Analysis of Deep Learning Models*. **IEEE S&P**, 2018
- [62] Arjun Nitin Bhagoji*, Warren He, Bo Li, Dawn Song. *Exploring the Space of Black-box Attacks on Deep Neural Networks*, **ECCV**, 2018.
- [63] Chaowei Xiao*, Duizhi Deng*, Bo Li, Fisher Yu, Mingyan Liu and Dawn Song. *Characterizing Adversarial Examples Based on Spatial Consistency Information for Semantic Segmentation*, **ECCV**, 2018
- [64] Lei Ma, Felix Juefei-Xu, Fuyuan Zhang, Jiyuan Sun, Minhui Xue, Bo Li, Chunyang Chen, Ting Su, Li Li, Yang Liu, Jianjun Zhao, and Yadong Wang. *DeepGauge: Multi-Granularity Testing Criteria for Deep Learning Systems*. **ASE**, 2018. (Distinguished Paper Award)
- [65] Chaowei Xiao*, Armin Sarabi, Yang Liu, Bo Li, Mingyan Liu, Tudor Dumitras. *From Patching Delays to Infection Symptoms: Using Risk Profiles for an Early Discovery of Vulnerabilities Exploited in the Wild*. **USENIX Security**, 2018
- [66] Chaowei Xiao*, Bo Li, Jun-Yan Zhu, Warren He, Mingyan Liu, Dawn Song. *Generating Adversarial Examples with Adversarial Networks*. **IJCAI**, 2018
- [67] Matthew Jagielski, Alina Oprea, Battista Biggio, Chang Liu, Cristina Nita-Rotaru, Bo Li. *Manipulating Machine Learning: Poisoning Attacks and Countermeasures for Regression Learning*. **IEEE S&P**, 2018

- [68] Ivan Evtimov, Kevin Eykholt*, Earlence Fernandes, Tadayoshi Kohno, Bo Li, Atul Prakash, Amir Rahmati, Chaowei Xiao, Dawn Song. *Robust Physical-World Attacks on Deep Learning Visual Classification*. **CVPR**, 2018
- [69] Xingjun Ma*, Bo Li, Yisen Wang, Sarah M. Erfani, Sudanthi Wijewickrema, Grant Schoenebeck, Dawn Song, Michael E. Houle, and James Bailey. *Characterizing Adversarial Subspaces Using Local Intrinsic Dimensionality*. **ICLR**, 2018. (Oral presentation).
- [70] Warren He, Bo Li and Dawn Song. *Decision Boundary Analysis of Adversarial Examples*. **ICLR**, 2018.
- [71] Chaowei Xiao*, Jun-Yan Zhu, Bo Li, Warren He, Mingyan Liu, and Dawn Song. *Spatially Transformed Adversarial Examples*. **ICLR**, 2018.
- [72] Xiaojie Wang*, Jianzhong Qi, Ramamohanarao Kotagiri, Yu Sun, Bo Li, and Rui Zhang. *A Joint Optimization Approach for Personalized Recommendation Diversification*. **PAKDD**, 2018.
- [73] Chao Yan*, Bo Li, Yevgeniy Vorobeychik, Aron Laszka, Daniel Fabbri, Bradley Malin. *Get Your Workload in Order: Game Theoretic Prioritization of Database Auditing*. **ICDE**, 2018.
- [74] Ruoxi Jia*, Ioannis Konstantakopoulos, Bo Li, Dawn Song, Costas J. Spanos. *Poisoning Attacks on Data-Driven Utility Learning in Games*. **ACC**, 2018.
- [75] Qingrong Chen*, Minhui Xue, Chong Xiang, Bo Li, Haizhong Zheng, and Haojin Zhu. *Do We Need Original Data for Training? Toward Designing Privacy-Preserving Machine Learning*. **DLSRF**, 2017. (**Research Forum Award**)
- [76] Lei Huang*, Xianglong Liu, Bo Lang, Adams Wei Yu, Bo Li. *Orthogonal Weight Normalization: Solution to Optimization over Multiple Dependent Stiefel Manifolds in Deep Neural Networks*. **AAAI**, 2017.
- [77] Chang Liu, Bo Li, Yevgeniy Vorobeychik, and Alina Oprea. *Robust Linear Regression Against Training Data Poisoning*. **AISec**, 2017. (**Best Paper Award**).
- [78] Ivan Evtimov, Kevin Eykholt*, Earlence Fernandes, Bo Li, Amir Rahmati, Chaowei Xiao, Atul Prakash, Tadayoshi Kohno, Dawn Song (alphabetical order) . *Robust physical-world attacks on deep learning visual classification*. In IEEE Conference on Computer Vision and Pattern Recognition, **CVPR** 2018. *Press: IEEE Spectrum, Yahoo News, Wired, Engadget, Telegraph, Car and Driver, CNET, Digital Trends, SCMagazine, Schneier on Security, Ars Technica, Fortune*
- [79] Bo Li, Kevin Roundy, Chris Gates and Yevgeniy Vorobeychik. Large-scale identification of malicious singleton files. In *ACM Conference on Data and Application Security and Privacy (CODASPY)*. Arizona, USA, 2017.
- [80] Bo Li, Yevgeniy Vorobeychik, Muqun Li, and Bradley Malin. Sanitizing Large-Scale Medical Records Before Publishing. In *AAAI Symposium on AI for Social Good (AISOC)*. Stanford University, 2017.
- [81] Jie Gao, Bo Li, Grant Schoenebeck and Fang-Yi Yu. Engineering Agreement: The Naming Game with Asymmetric and Heterogeneous Agents, In *Proceedings of the 30th International Conference on Artificial Intelligence (AAAI)*. San Francisco, USA, 2017.
- [82] Bo Li, Yining Wang, Aarti Singh, and Yevgeniy Vorobeychik. Data Poisoning Attacks on Factorization-based Collaborative Filtering, In *Proceedings of the Neural Information Processing Systems (NIPS)*. Barcelona, Spain, 2016.

- [83] Bo Li. Secure Learning in Adversarial Environments. *In Proceedings of ACM Career Network and Conference (SIGAI CNC)*. Boston, USA, 2016.
- [84] Chao Yan, You Chen, Bo Li, David Liebovitz, and Bradley Malin. Learning Clinical Workflows to Identify Subgroups of Heart Failure Patients. *In Proceedings of the American Medical Informatics Association (AMIA)*. Chicago, USA, 2016.
- [85] Liyiming Ke, Bo Li, and Yevgeniy Vorobeychik. Behavioral Experiments in Email Filter Evasion. *In Proceedings of the 30th International Conference on Artificial Intelligence (AAAI)*. Phoenix, AZ, USA, 2016.
- [86] Bo Li, Yevgeniy Vorobeychik, Muqun Li, Bradley Malin. Scalable Iterative Classification for Sanitizing Large-Scale Datasets. *IEEE Transactions on Knowledge and Data Engineering (TKDE)*, Oct. 2016.
- [87] Bo Li, Yevgeniy Vorobeychik, Muqun Li, and Bradley Malin. Iterative Classification for Sanitizing Large-Scale Datasets. *In Proceedings of the IEEE International Conference on Data Mining (ICDM)*. Atlantic City, NJ, USA, 2015.
- [88] Bo Li. Secure Learning and Mining in Adversarial Environments. *In Proceedings of the IEEE International Conference on Data Mining (ICDM) Phd Forum*. Atlantic City, NJ, USA, 2015.
- [89] Bo Li and Yevgeniy Vorobeychik. Scalable Optimization of Randomized Operational Decisions in Adversarial Classification Settings. *In Proceedings of the 18th International Conference on Artificial Intelligence and Statistics (AISTATS)*. San Diego, CA, USA, 2015.
- [90] Jung-San Lee, Bo Li, and Bo-Ruei Huang. Oblivious Transfer System for Information Concealing and Recovery in Cloud Computing. *Journal of Internet Technology*, vol. 20, pp. 132-145, March 2015.
- [91] Jung-San Lee, Kai-Rui Wen, and Bo Li. Renovating Contaminative Image Archives based on Patch Propagation and Adaptive Confidence Collation. *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 14, pp. 112-134, April 2015.
- [92] Muqun Li, David Carrell, John Aberdeen, Lynette Hirschman, Bo Li, Yevgeniy Vorobeychik, and B. Malin. Optimizing Annotation Resources for Natural Language De-identification via a Game Theoretic Framework. *Journal of Biomedical Informatics*, vol. 30, pp.142-179, October 2015.
- [93] Bo Li and Yevgeniy Vorobeychik. Feature Cross-substitution in Adversarial Classification. *In Proceedings of the Neural Information Processing Systems (NIPS)*. Quebec, Canada, 2014.
- [94] Yihua Wang, Jung-San Lee, and Bo Li. A Hybrid Illumination Refinement Technique Based on Double Niblack Thresholds. *In Proceedings of the International Conference on Communication and Information Systems (ICCIS)*. Roma, Italy, 2015.
- [95] Jung-San Lee, K. J. Wei, B. R. Huang, and Bo Li. Note on an Anonymous Multi-server Authenticated Key Agreement Scheme Based on Trust Computing using Smart Cards and Biometrics. *In Proceedings of the Cryptography and Information Security Conference (CISC)*. Kaohsiung, Taiwan, 2015.
- [96] Mary Ellen Koran, Bo Li, Neda Jahanshad, Tricia A. Thornton-Wells, David C. Glahn, Paul M. Thompson, John Blangero, Thomas E. Nichols, Peter Kochunov, and Bennett A. Landman. The Impact of Family Structure and Common Environment on Heritability Estimation for Neuroimaging Genetics Studies using SOLAR. *In*

- Proceedings of the International Imaging Genetics Conference (GC)*. Irvine, CA, USA, 2014.
- [97] Yevgeniy Vorobeychik and Bo Li. Optimal Randomized Classification in Adversarial Settings. In *Proceedings of the International Joint Conference on Autonomous Agents and Multiagent Systems (AAMAS)*. Paris, France, 2014.
 - [98] Mary Ellen Koran, Bo Li, Neda Jahanshad, Tricia A. Thornton-Wells, David C. Glahn, Peter. M. Thompson, John Blangero, Thomas E. Nichols, Peter Kochunov, and Bennett A. Landman. Neuroimaging Genetics Study Design: A Comparison of the OpenMX and SOLAR Software Packages for Estimating Heritability of Neuroimaging Traits. In *Proceedings of the International Imaging Genetics Conference*. Irvine (GC), CA, USA, 2014.
 - [99] Mary Ellen Koran, Bo Li, Neda Jahanshad, Tricia A. Thornton-Wells, David C. Glahn, Paul M. Thompson, John Blangero, Thomas E. Nichols, Peter Kochunov, and Bennett A. Landman. On Study Design in Neuroimaging Heritability Analyses. In *Proceedings of the SPIE Medical Imaging Conference (SPIE)*. San Diego, CA, USA, 2014.
 - [100] Zhoubing Xu, Bo Li, Swetasudha Panda, Andrew J. Asman, Kristen L. Merkle, Peter L. Shanahan, Richard G. Abramson, and Bennett A. Landman. Shape-Constrained Multi-Atlas Segmentation of Spleen in CT. In *Proceedings of the SPIE Medical Imaging Conference (SPIE)*. San Diego, CA, USA, 2014.
 - [101] Jung-San Lee, K. J. Wei, and Bo Li. On the Security of Two ECC-based Authentication Mechanisms for Confirming Authorization. In *Proceedings of the Cryptography-20 years of Cryptography in Taiwan, McGraw-Hill International Enterprises, LLC*. Taiwan, 2014.
 - [102] Jung-San Lee, and Bo Li. Self-recognized Watermarking Technique with Resistance to Large Scale Cropping. *IEEE Transactions on Multimedia*, vol. 21, pp. 60-73, March 2014.
 - [103] Jung-San Lee, Wei-Chiang Kao, and Bo Li. Aryabhata Remainder Theorem-Based Non-Iterative Electronic Lottery Mechanism with Robustness. *IET Information Security*, vol. 7, pp. 172-182, May 2013.
 - [104] Jung-San Lee, and Bo Li. Note on Authentication Protocol using an Identifier in an Ad hoc Network Environment. *Mathematical and Computer Modeling*, vol. 57, pp. 1572-1575, March 2013.
 - [105] Qiufeng Lin, Zhoubing Xu, Bo Li, Rebecca Baucom, Benjamin Poulouse, Bennett A. Landman, and Robert E. Bodenheimer. Immersive Virtual Reality for Visualization of Abdominal CT. In *Proceedings of the SPIE Medical Imaging Conference (SPIE)*. Orlando, FL, USA, 2013.
 - [106] Bo Li, Swetasudha Panda, Zhoubing Xu, Andrew J. Asman, Peter L. Shanahan, Richard G. Abramson, and Bennett A. Landman. Regression Forest Region Recognition Enhances Multi-atlas Spleen Labeling. In *Proceedings of the Medical Image Computing and Computer Assisted Intervention (MICCAI)*. Nagoya, Japan, 2013.
 - [107] Bo Li, B. S. Caffo, A. S Gokhale, and B. A. Landman. Medical Image Processing in the Cloud. In *Proceedings of the International Imaging Genetics Conference (GC)*. Irvine, CA, USA, 2013.

- [108] Bo Li, Peter Volgyesi, and Akos Ledeczki. Rapid Prototyping of Image Processing Workflows on Massively Parallel Architectures. *In Proceedings of WISES 2012 (WISES)*. Carinthia, Austria, 2012.
- [109] Jung-San Lee, Bo Li, and Hsin-Yi Tsai. An Effective Preprocessor for Enhancing Edge Detection in Speckle Images. *In Proceedings of the 8th International Conference on Information Science and Digital Content Technology (ICIDT)*. Jeju, Korea, 2012.
- [110] Bo Li, Frederick Bryan, and Bennett A. Landman. Next Generation of the JAVA Image Science Toolkit (JIST) Visualization and Validation. *InSight Journal*, vol. 12, pp. 23-34, February 2012.
- [111] Chien-Chang Yang, Kup-jui Wei, Bo Li and Jung-San Lee. HORME: Hierarchical-Object-Relational Medical Management for Electronic Record. *Security and Communication Networks*, vol. 17, pp. 53-65, November 2012.
- [112] F. P. Lin, Bo Li and Jung-San Lee. A Neoteric Progressive Scheme for Visual Secret Sharing. *In Proceedings of the First National Conference on Web Intelligence and Applications (CWIA)*. Kaohsiung, Taiwan, 2011.
- [113] Bo Li and Jung-San Lee. A wally Watermarking Scheme with Blind Extraction Based on Structured Encoding System. *In Proceedings of International Conference on 2nd World Congress on Computer Science and Information Engineering (CSIE)*. Changchun, China, 2011.
- [114] Bo Li, Jung-San Lee, and Chin-Chen Chang. Providing Anonymous Channels and Mutual Authentication in GSM. *In Proceedings of International Conference on the Fourth International Conference on Genetic and Evolutionary Computing (WCECS)*. Shenzhen, China, 2010.
- [115] Chin-Chen Chang, Bo Li and Jung-San Lee. Secret Sharing Using Visual Cryptography. *Journal of Electronic Science and Technology*, vol. 8, pp. 289-299, June 2010.
- [116] Bo Li and Jung-San Lee. A Cryptographic Alternative for Preserving PHI in Compliance with HIPAA Privacy/Security Regulations. *Journal of Computers*, vol. 21, pp. 27-36, March 2010.
- [117] Jung-San Lee, Bo Li, and Yung-Chen Chou. A Brand-new Mobile Value-added Service: M-Check. *In Proceedings of International Conference on Networked Computing, Advanced Information Management and Digital Content and Multimedia Technologies (AICIT)*. Seoul, Korea, 2009.

PATENTS

- Systems and Methods for Identifying Suspicious Singleton Files Using Correlation Predictors (10073983. w/ Symantec)
- Neural Energy Saver: Model-Agnostic Input Transformation Learning for Adjusting Accuracy-Energy Tradeoff in Neural Networks (96665195. w/ IBM)
- State-Augmented Reinforcement Learning (P201908998US01. w/ IBM)
- Model-agnostic Input Transformation for Neural Networks (P202102795US01. w/ IBM)

PROFESSIONAL ACTIVITIES

Tutorials

- AAMAS Conference, May, 2019 (Adversarial Machine Learning)

- AAAI Conference on Artificial Intelligence, February, 2018/2019 (Adversarial Machine Learning, together with Yevgeniy Vorobeychik and Dawn Song; > 300 registered attendees)
- IJCAI Conference, January, 2020 (Machine Learning and Game Theory)

Workshops

(I served as the lead-organizer for the following workshops.)

- Workshop on The Art of Robustness: Devil and Angel in Adversarial Machine Learning, CVPR 2022.
- Workshop on A Blessing in Disguise: The Prospects and Perils of Adversarial Machine Learning, ICML 2021.
- Workshop on Socially Responsible Machine Learning, ICML 2021.
- Workshop on Adversarial Machine Learning in Real-World Computer Vision Systems and Online Challenges , CVPR 2021/2022.
- Workshop on Safety & Security of Deep Learning, IJCAI 2021.
- Workshop on Security and Safety in Machine Learning Systems, ICLR 2021.
- Workshop on Adversarial Learning Methods for Machine Learning and Data Mining, KDD 2019/ 2020/ 2021.
- Workshop on Adversarial Machine Learning in Computer Vision, CVPR 2020.
- Workshop on Data Curation and Security, NeurIPS 2020.
- Workshop on Adversarial Robustness in the Real World, ECCV 2020.
- Workshop on Security and Privacy of Machine Learning, ICML 2019.
- Workshop on Socially Responsible Machine Learning, ICML 2019.
- Workshop on Adversarial Machine Learning in Real-World Computer Vision Systems, CVPR 2019.
- Workshop on Machine Learning and Computer Security, NeurIPS 2017.

Program Committee

- 2021: NSF panel on Data Science
- 2021: NeuIPS, ICML, ICLR, IEEE S&P, CCS
- 2020: NeuIPS, ICML, ICLR, IEEE S&P, CCS, Usenix Security, NDSS
- 2019: NeuIPS, ICML, AISTATS, ICLR, IEEE S&P, Usenix Security, CCS, NDSS
- 2018: NeuIPS, ICML, AISTATS, IEEE S&P, CCS, NDSS
- 2017: NeuIPS, ICML, ICLR, IEEE S&P, Usenix Security, NDSS
- 2016: NeuIPS, ICML, ICLR, IEEE S&P, NDSS
- 2015: NeuIPS, ICML, CVPR, AISTATS, AAAI

Area Chair

- 2021: CVPR, ICCV, ICML, NeurIPS
- 2020: CVPR, AAAI, ICML, NeurIPS, IJCAI
- 2019: CVPR, AAAI, NeurIPS, IJCAI
- 2018: CVPR, AAAI, IJCAI
- 2017: CVPR, AAAI, AAMAS
- 2016: AAAI, AAMAS

Journal Reviewer/Editorial Board

- Reviewer for IEEE Transactions on Multi-media; IEEE Transactions on Internet Computing; JMLR
- Editorial Board for Journal of Surveillance, Security and Safety; JAAIML; IEEE Signal Processing Society

Speeches & Seminars (Selected)

- 02/2022 UK Security and Privacy Seminar
- 12/2021 Invited talk at WiML 2021

- 11/2021 Invited talk at Amazon
- 11/2021 Annual Workshop on Federated Learning and Analytics hosted by Google
- 10/2021 Invited talk at the AI seminar at CMU
- 10/2021 4th World Laureates Forum
- 10/2021 Keynote at the ACM Multimedia Workshop on Trustworthy AI
- 10/2021 Invited talk at the Center for Artificial Intelligence Innovation (CAII)
- 10/2021 Invited talk at NIO
- 09/2021 Colloquium Talk at McGill University
- 07/2021 Invited talk at EmTech Europe 2021
- 06/2021 Keynote at the Workshop on Autonomous Driving (WAD) at CVPR
- 05/2021 Invited talk at Intel
- 05/2021 Invited talk at the RobustML Workshop at ICLR
- 04/2021 Invited talk at Facebook
- 04/2021 Guest lecture in Provably Safe Robotics at CMU
- 04/2021 DARPA ISAT Study on Designing and Resisting AI Cyber User-in-the-Loop Attacks (DRACULA)
- 03/2021 Guest lecture on Safety for Autonomous Driving AI at CMU
- 03/2021 C3SR talk at UIUC-IBM AI series
- 01/2021 BAAI Seminar
- 11/2020 CSAIL Security Seminar Series, MIT
- 10/2020 Innovators' Think Tank - AI and Quantum, MIT TR-35
- 10/2020 SPAI worksho in AsiaCCS
- 10/2020 Guest lecture at University of Illinois at Urbana-Champaign
- 09/2020 TEDxGateway
- 08/2020 ECCV workshop on Adversarial Learning in the Real World
- 07/2020 Workshop on Socially Responsible Machine Learning at ICML
- 07/2020 International Workshop on Verification of Neural Networks (VNN20)
- 06/2020 CVPR 2019 Workshop on Autonomous Driving
- 04/2020 Guest lecture in Safe Robotics at CMU
- 03/2020 C3SR talk at UIUC-IBM AI series
- 10/2019 Open Data Science Conference (ODSC)
- 08/2019 Waterloo ML+Security+Verification workshop
- 09/2019 Distinguished Seminar Series in Cylab, CMU
- 07/2019 IBM AI Research Week Workshop on Foundations of Safe Learning
- 06/2019 CVPR 2019 Workshop on Autonomous Driving
- 06/2019 Ant Financial ICML Expo Workshop
- 05/2019 International Conference AI and Adaptive Education
- 03/2019 Workshop on the implications of AI for cybersecurity, National Academies of Sciences, Engineering, and Medicine
- 09/2018 Keynote at GlobalSIP
- 06/2018 CVPR 2019 Workshop on Autonomous Driving
- 08/2018 GeekPwn, Las Vegas

PRESS

IEEE Spectrum, Yahoo News, Wired, Engadget, Telegraph, Car and Driver, CNET, Digital Trends, SCMagazine, Schneier on Security, Ars Technica, Fortune, Nature, Science

CURRENT STUDENTS AND POSTDOCS

- Jian Liu, (Postdoc)
- Huichen Li (Ph.D. student)
- Linyi Li, (Ph.D. student)
- Boxin Wang, (Ph.D. student, **Yunni & Maxine Pao Memorial Fellowship, Top-3 Intern Award at MSR 2020**)

- Zhuoling Yang, (Ph.D. student)
- Haoxiang Wang, (Ph.D. student)
- Mantas Mazeika, (Ph.D. student)
- Fan Wu, (Ph.D. student)
- Chulin Xie, (Ph.D. student)
- Hyun Ho Son, (Ph.D. student)
- Evelyn Ma, (Ph.D. student)
- Jiawei Zhang (MS student)
- Weijie Lyu (MS student)
- Zijian Huang (MS student)

ALUMNI

- Yunhui Long (PhD. Researcher at Facebook)
- Kevin Eykholt (PhD. Researcher at IBM)
- Chaowei Xiao (PhD. Assistant Professor at Arizona State University)
- Steven Liu (PhD. Twitter)
- Siwakorn Srisakaokul (MS. Securities and Exchange Commission)
- Zhe Xu (MS. Alibaba)
- Devansh Paresh Shah (MS. Amazon)
- Krishna Harsha Reddy Kothapalli (MS. Google)
- Haroun Habeeb (MS. Facebook)
- Rishika Agarwal (MS. Apple)

GRANTS AND OTHER FUNDING

Research Grant (2022-2027) co-Principal Investigator, “Robust and Resilient Autonomy for Advanced Air Mobility”, funded by NASA. (\$244,601)

Research Grant (2022) Principal Investigator, Sloan Award. (\$75,000)

Research Grant (2022-2023) Principal Investigator, “Robust and Private Learning on Graph Structured Data”, funded by Ebay. (\$115,000)

Research Grant (2021-2022) Principal Investigator, “Reinforcement Learning for a Resilient Electric Power System”, funded by C3.ai DTI Research Award. (\$75,000)

Research Grant (2020-2021) Principal Investigator, “Certifiably Private, Robust, and Explainable Federated Learning”, funded by the Facebook Research Award. (\$50,000)

Research Grant (2020-2021) Principal Investigator, “Trustworthy Machine Learning”, funded by the Lawrence Livermore National Laboratory. (\$110,000)

Research Grant (2021-2026) Principal Investigator, “DeepTrust: Enabling Robust Machine Learning with Exogenous Information”, funded by the NSF CARRER Award. (\$500,000)

Research Grant (2020-2021) Principal Investigator, “Trustworthy Distributed ML Systems”, funded by the Amazon Research Awards. (\$60,000)

Research Grant (2020-2021) Principal Investigator, “Robust Multimedia Recommendation”, funded by the Intel Research Rising Stars Award. (\$50,000)

Research Grant (2020-2021) Principal Investigator, “Robust Autonomous Driving System”, funded by the Baidu Research Award. (\$90,000)

Research Grant (2019-2022) co-Principal Investigator, “Rigorous Approaches for Scalable Privacy-preserving Deep Learning”, funded by the NSF FA. (\$208,000)

Research Grant (2019-2020) Principal Investigator, “Trustworthy Machine Learning”, funded by the Lawrence Livermore National Laboratory. (\$110,000)

Research Grant (2019-2021) co-Principal Investigator, “SHIELD: A Statistical Machine Learning Framework for Diversity Enabled Ensemble Robustness”, funded by the DARPA AIE. (\$300,000)

Research Grant (2019-2020) Principal Investigator, “Robust AutoML Pipeline”, funded by the Amazon Research Awards. (\$34,032)

Research Grant (2018-2019) Principal Investigator, “Adversarial Machine Machine Learning”, funded by the Lawrence Livermore National Laboratory. (\$98,497)

Research Grant (2018-2019) co-Principal Investigator, “LASER: Learning to Automate Social Engineering Resistance”, funded by the DARPA ASER. (\$189,628)